LECTURE 5

Q.M. Laws needed for Computation:

(1) Initialize a quantum state $|0\rangle$ (w.l.o.g.)

(2) Unitary transformation

(3) (Partial) Measurement. (in standard basis w.l.o.g.)

No cloning theorem

There is no quantum that clones an unknown qubit

Proof: Suppose there is!

$$\forall \phi \quad Q\,|\phi\rangle|0^n\rangle = |\phi\rangle|\phi\rangle|f(\phi)\rangle$$

$$\Rightarrow \quad Q\,|0\rangle\,|0^n\rangle = |0\rangle\,|0\rangle\,|f(0)\rangle$$

$$Q\,|1\rangle\,|0^n\rangle = |1\rangle\,|1\rangle\,|f(1)\rangle$$

By linearity. $Q\left(|+\rangle\,|0^n\rangle\right) = \frac{1}{\sqrt{2}}\left(|00\,f(0)\rangle + |11\,f(1)\rangle\right)$

$$\neq \quad \left(|++\,f(+)\rangle\right)$$

Quantum teleportation

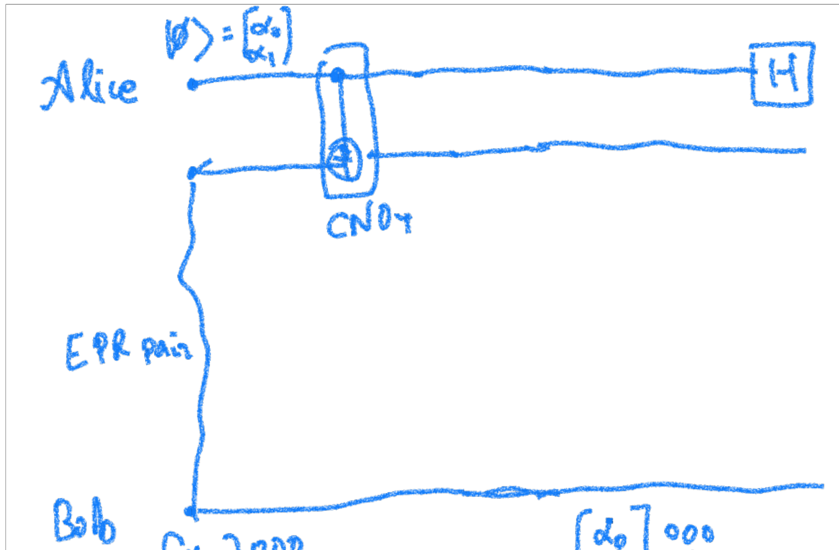
Alice $\sim\sim\sim$ Bob (share an EPR pair)

$\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$

Bob goes far away (while maintaining the EPR pair)

Alice has another qubit $\phi = \alpha_0|0\rangle + \alpha_1|1\rangle = \begin{bmatrix}\alpha_0 \\ \alpha_1\end{bmatrix}$

Alice wants to send to Bob ϕ by just sending Bob a short message



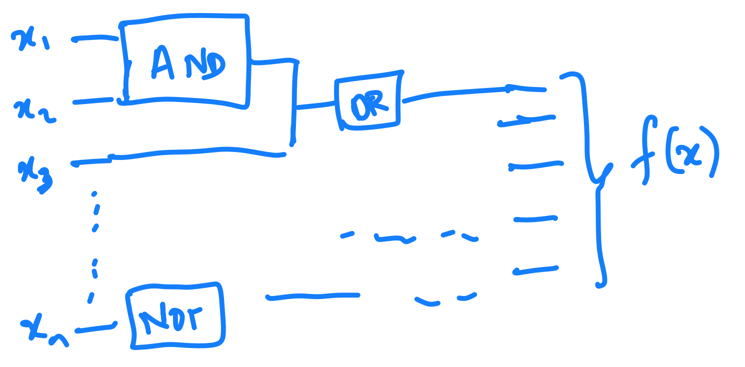
$$\begin{aligned}&\frac{1}{2}|00\rangle \otimes (\alpha_0|0\rangle + \alpha_1|1\rangle)\\ &+ \frac{1}{2}|01\rangle \otimes (\alpha_1|0\rangle + \alpha_0|1\rangle)\\ &+ \frac{1}{2}|10\rangle \otimes (\alpha_0|0\rangle - \alpha_1|1\rangle)\\ &+ \frac{1}{2}|11\rangle \otimes (-\alpha_0|0\rangle + \alpha_1|1\rangle)\end{aligned}$$

$$\begin{bmatrix}\alpha_0\\0\\0\\\alpha_0\\\alpha_1\\0\\0\\\alpha_1\end{bmatrix}\begin{array}{l}000\\001\\010\\011\\100\\101\\110\\111\end{array} \qquad \frac{1}{\sqrt{2}}\begin{bmatrix}\alpha_0\\0\\0\\\alpha_0\\0\\\alpha_1\\\alpha_1\\0\end{bmatrix}\begin{array}{l}000\\001\\010\\011\\100\\101\\110\\111\end{array} \qquad \frac{1}{2}\begin{bmatrix}\alpha_0\\\alpha_1\\\alpha_1\\\alpha_0\\\alpha_0\\-\alpha_1\\-\alpha_1\\\alpha_0\end{bmatrix}\begin{array}{l}000\\001\\010\\011\\100\\101\\110\\111\end{array}$$

Basics of quantum computation :

Classical computation

Circuit model of computation for computing $f: \{0,1\}^n \to \{0,1\}^m$



Theorem (70's)

For any TM that computes $f : \{0,1\}^n \to \{0,1\}^m$ and runs in time $T(n)$, there exists a circuit $C : \{0,1\}^n \to \{0,1\}^m$ with $O(T(n) \log T(n))$ gates that computes f.

# of gates = size of circuit

Probabilistic circuit

Add another gate

$\boxed{\text{FLIP}} \leftarrow$ 0/1 with probability ½ each.

We say that a prob. circuit C computes

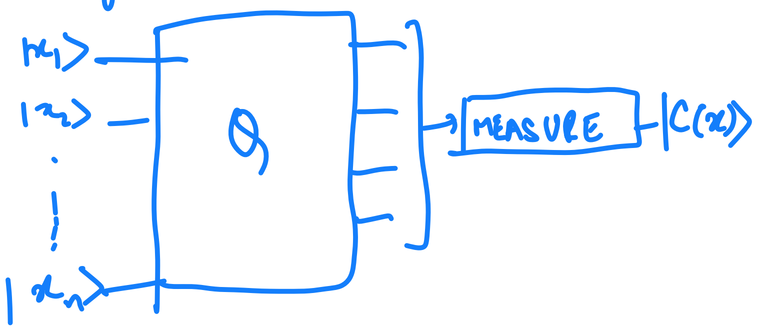
$F : \{0,1\}^n \rightarrow \{0,1\}^m$ if $\forall x$ $\Pr_{r}\left[C(x) = F(x)\right] \geq 0.99$

FLIP gates

Quantum circuit to compute $F : \{0,1\}^n \to \{0,1\}^m$

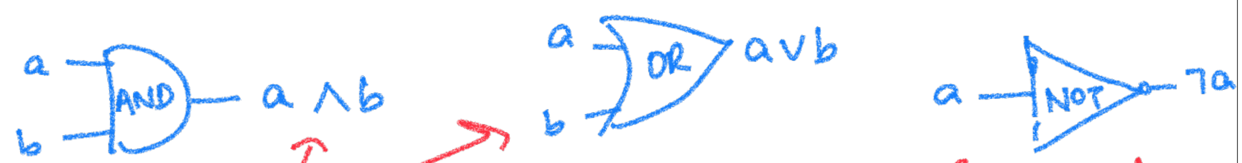
AND/OR/NOT gates replaced by unitary transformation on 1/2/3 qubits

given $x_1, \ldots\ldots, x_n \in \{0,1\}^n$



$$\Pr[F(x) = C(x)] \geq 0.99$$

- How to build quantum circuits that implement classical circuits
- Quantum operations are reversible
- What about



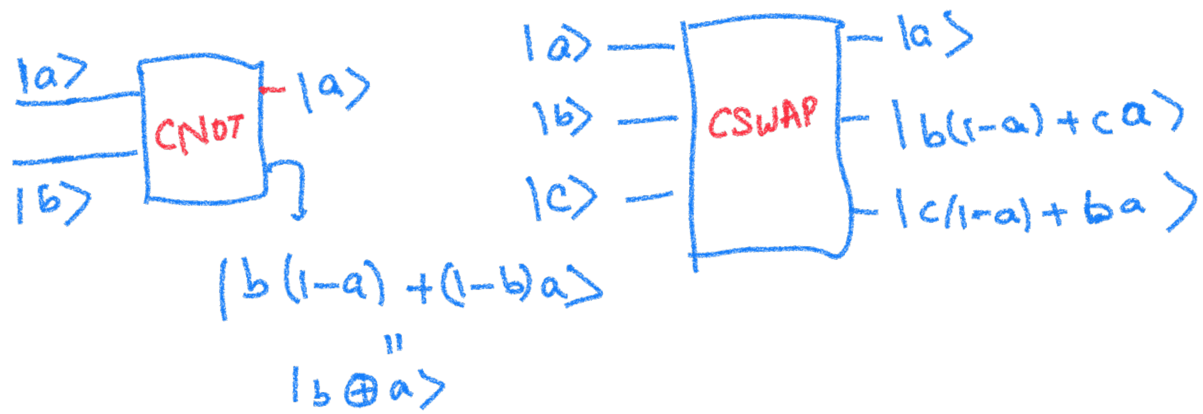
Not Reversible

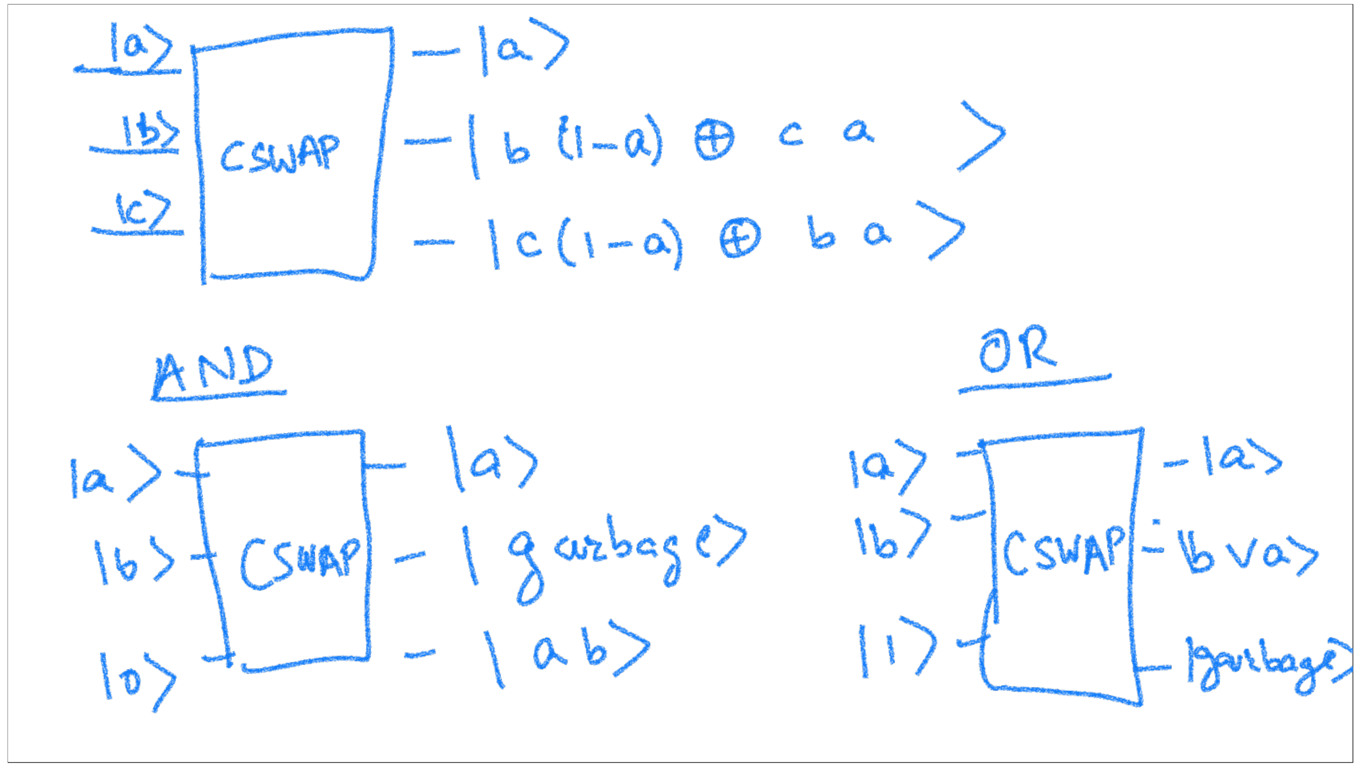
Reversible

CAN WE IMPLEMENT THESE VIA A REVERSIBLE QUANTUM CKT.

- Lets look at non-trivial quantum gates for which output qubits are "bits" when input qubits are "bits"

Assume $|a\rangle, |b\rangle, |c\rangle$ are either $|0\rangle$ or $|1\rangle$



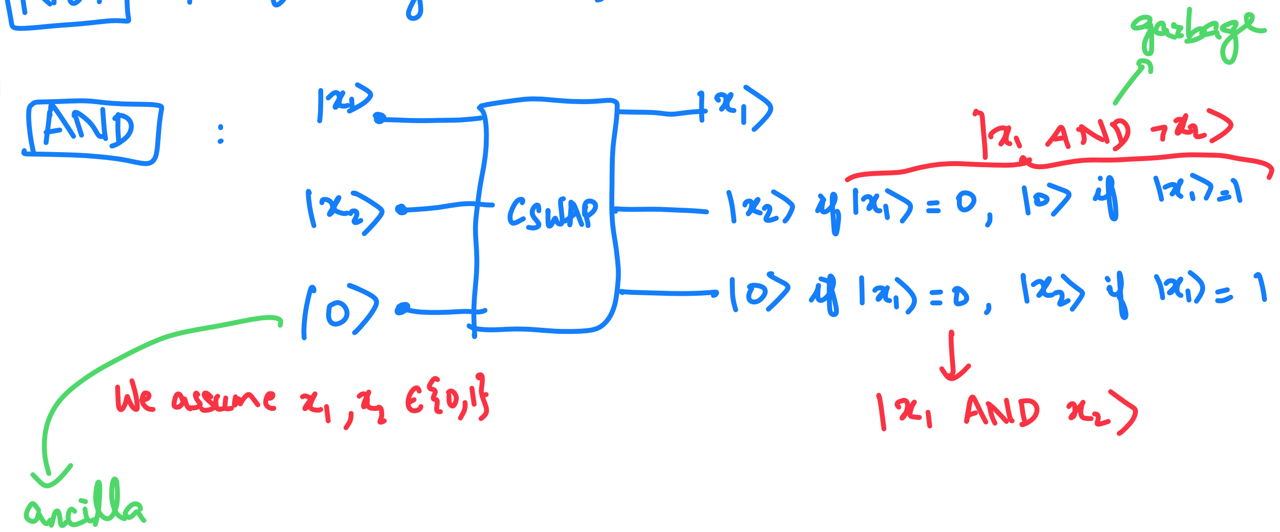


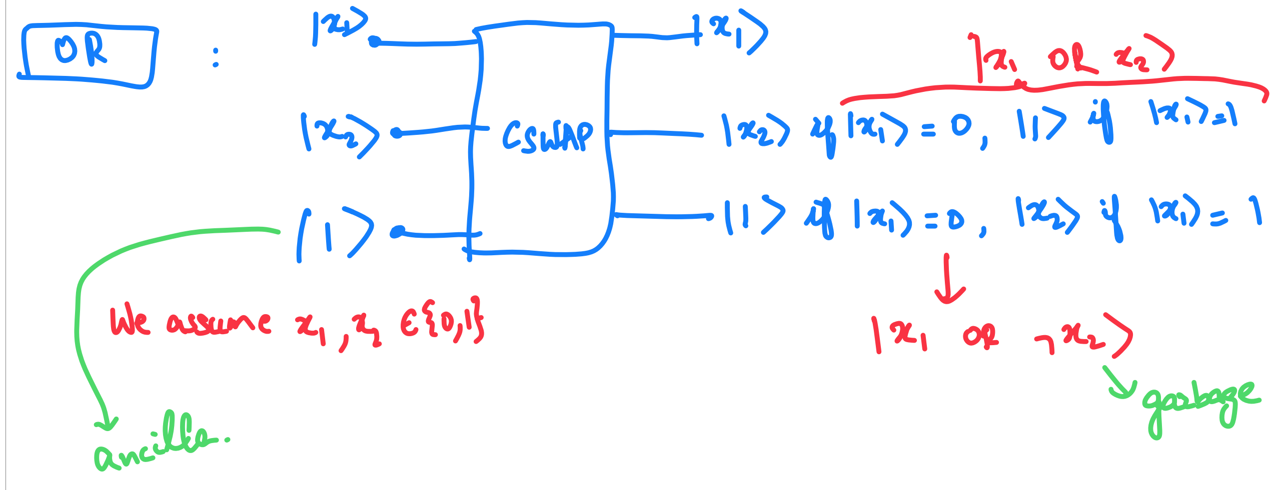
|a⟩
|b⟩
|c⟩
CSWAP
|a⟩
|b(1−a) ⊕ c a⟩
|c(1−a) ⊕ b a⟩
AND
|a⟩
|b⟩
|0⟩
CSWAP
|a⟩
|garbage⟩
|ab⟩
OR
|a⟩
|b⟩
|1⟩
CSWAP
|a⟩
|b ∨ a⟩
|garbage⟩

Implementing classical gates via unitary transformation

$\boxed{\text{NOT}}$: Unitary transformation.

$\boxed{\text{AND}}$:





OR :
|x₁⟩
|x₂⟩
|1⟩
CSWAP
|x₁⟩
|x₂⟩ if |x₁⟩ = 0, |1⟩ if |x₁⟩ = 1
|1⟩ if |x₁⟩ = 0, |x₂⟩ if |x₁⟩ = 1
|x₁ OR x₂⟩
|x₁ OR ¬x₂⟩
garbage
We assume x₁, x₂ ∈ {0,1}
ancilla.

OTHER OPERATIONS

Quantum?

FLIP — b

↑ uniformly random bit

$|0\rangle$ — H — Measure

a — Duplicate — a, a

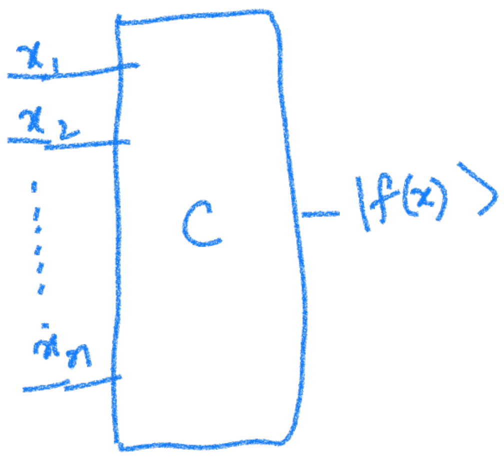
$|a\rangle$ — CNOT — $|a\rangle$

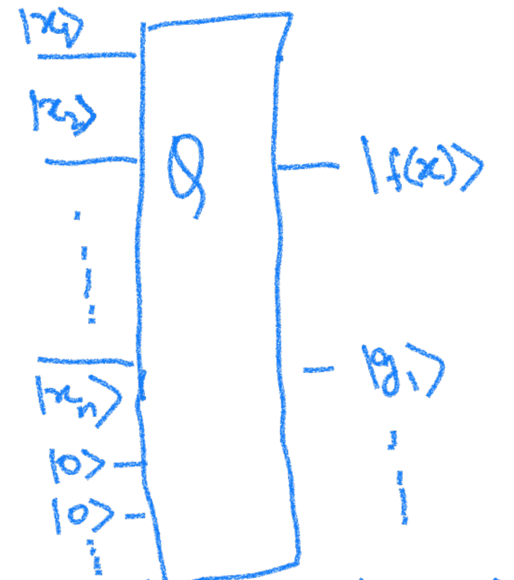
$|0\rangle$ — CNOT — $|a \oplus 0\rangle$

$= |a\rangle$

Is the no cloning thm false?

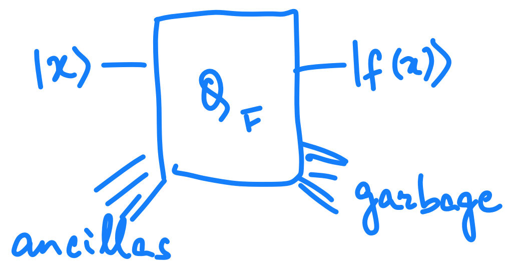
NO, works only for a = 0/1





$$Q\Big(|\text{Input}\rangle \otimes \underset{\substack{\uparrow \\ \text{ancilla}}}{|0^a\rangle}\Big) = |\text{Output}\rangle \otimes \underset{\substack{\uparrow \\ \text{garbage}}}{|\ldots\rangle}$$

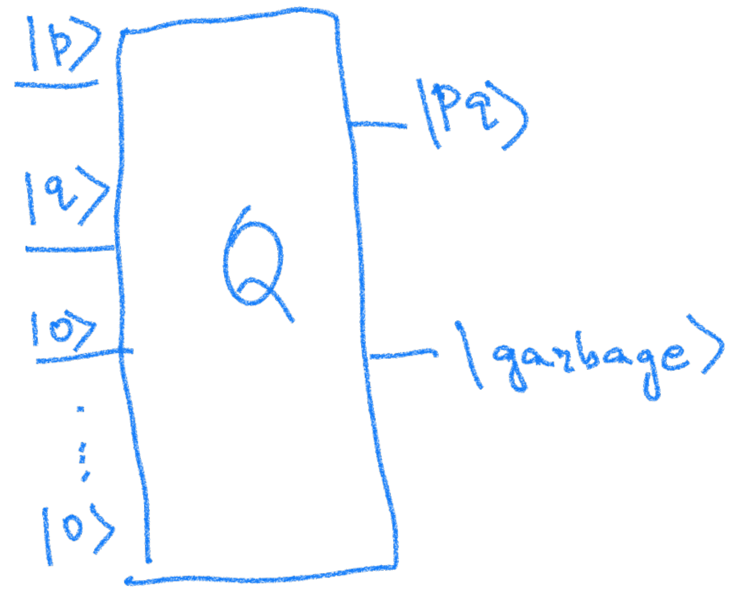
Quantum Circuit for $F : \{0,1\}^n \to \{0,1\}^m$



$$n + a = m + g$$

Q_F invertible unitary transformation implemented as a combination of elementary unitary transformations on 1/2/3 qubits.

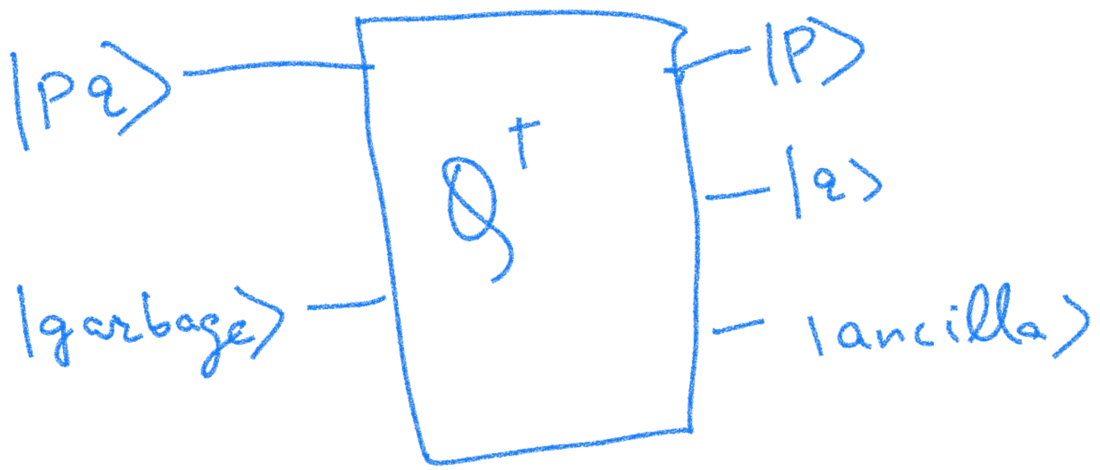
- So we can at least implement any classical computation via a quantum circuit
- Any classical computation can be done reversibly
- So, can we solve Integer factorization?



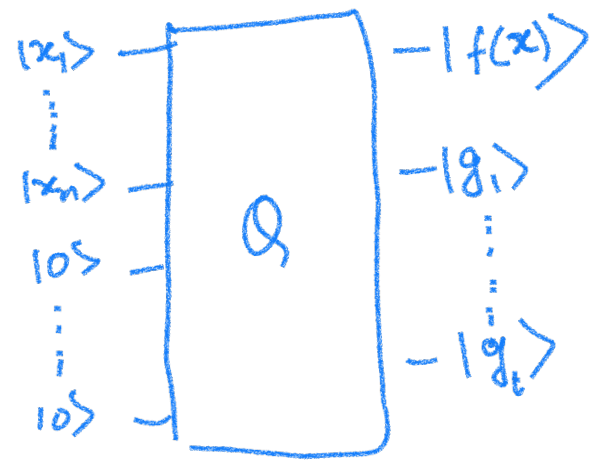
$N = pq$

To factor, we need "garbage"

If we know garbage qubits we could factor

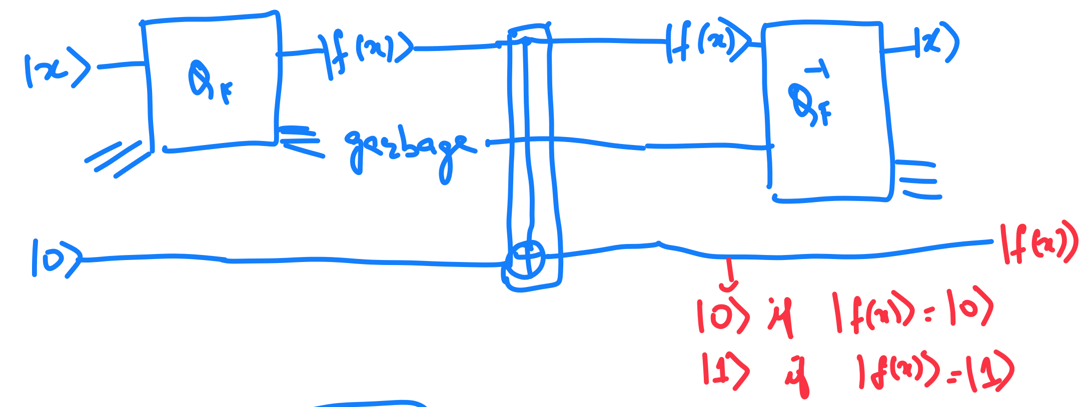


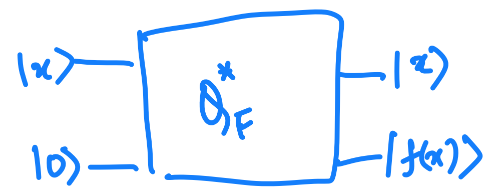
UNCOMPUTING GARBAGE



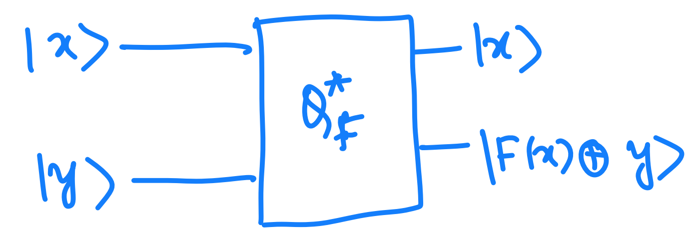
$$|x\rangle \otimes |0^a\rangle \longrightarrow f(x) \otimes |g\rangle$$

Uncomputing garbage $F : \{0,1\}^n \to \{0,1\}$





More generally, for $F: \{0,1\}^n \to \{0,1\}^m$



$y \in \{0,1\}^m$

We will not always write the ancilla qubits $|0^a\rangle$

We will often care about boolean functions

$$F : \{0,1\}^n \to \{0,1\}$$ $\to$ output is 1-bit.

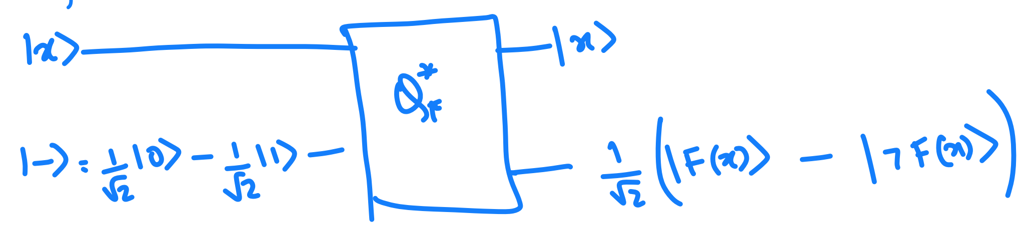
2^{2^n} possible functions.

Sometimes useful to work with $f : \{0,1\}^n \to \{-1,1\}$

$$f(x) = (-1)^{F(x)}$$

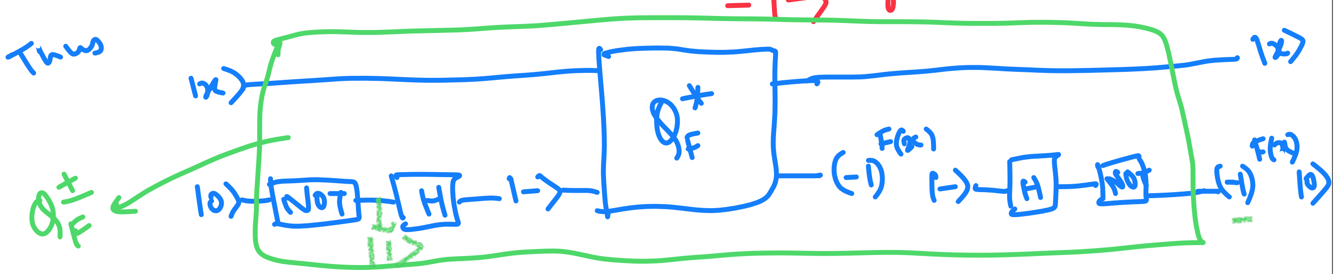
$$\begin{aligned} f(x) &= 1 \text{ if } F(x) = 0 \\ &= -1 \text{ if } F(x) = 1 \end{aligned}$$

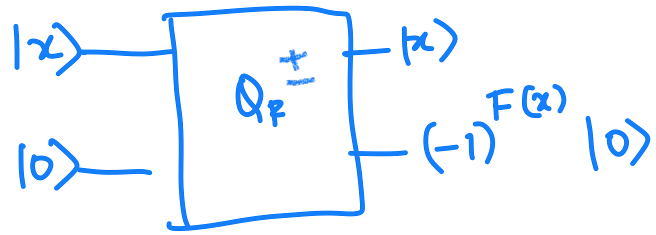
Useful trick : $Q_F^{\pm}$ for $F : \{0,1\}^n \to \{0,1\}$



$$\left.\begin{array}{ll} |-\rangle & \text{if } F(x) = 0 \\ -|-\rangle & \text{if } F(x) = 1 \end{array}\right\} (-1)^{F(x)}|-\rangle$$

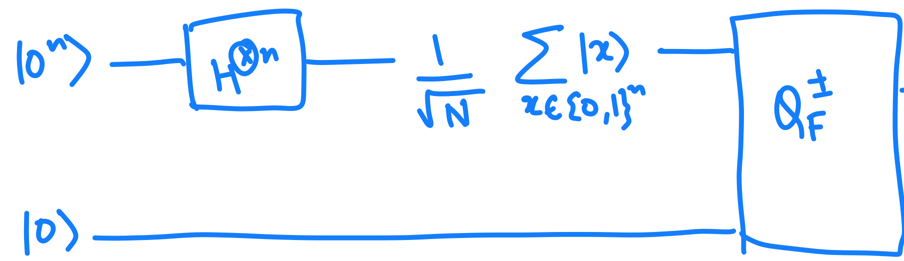
Thus





$$(-1)^{F(x)} |x\rangle \otimes |0\rangle$$

Useful idea (A single application of $Q_F^{\pm}$ creates a state which has info about $F(x)$ for all x.)

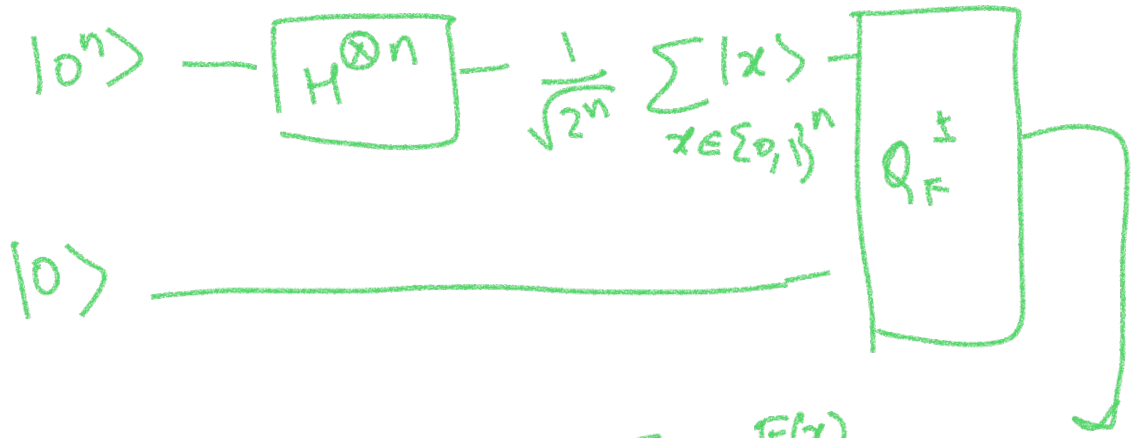


$$\frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} |x\rangle$$

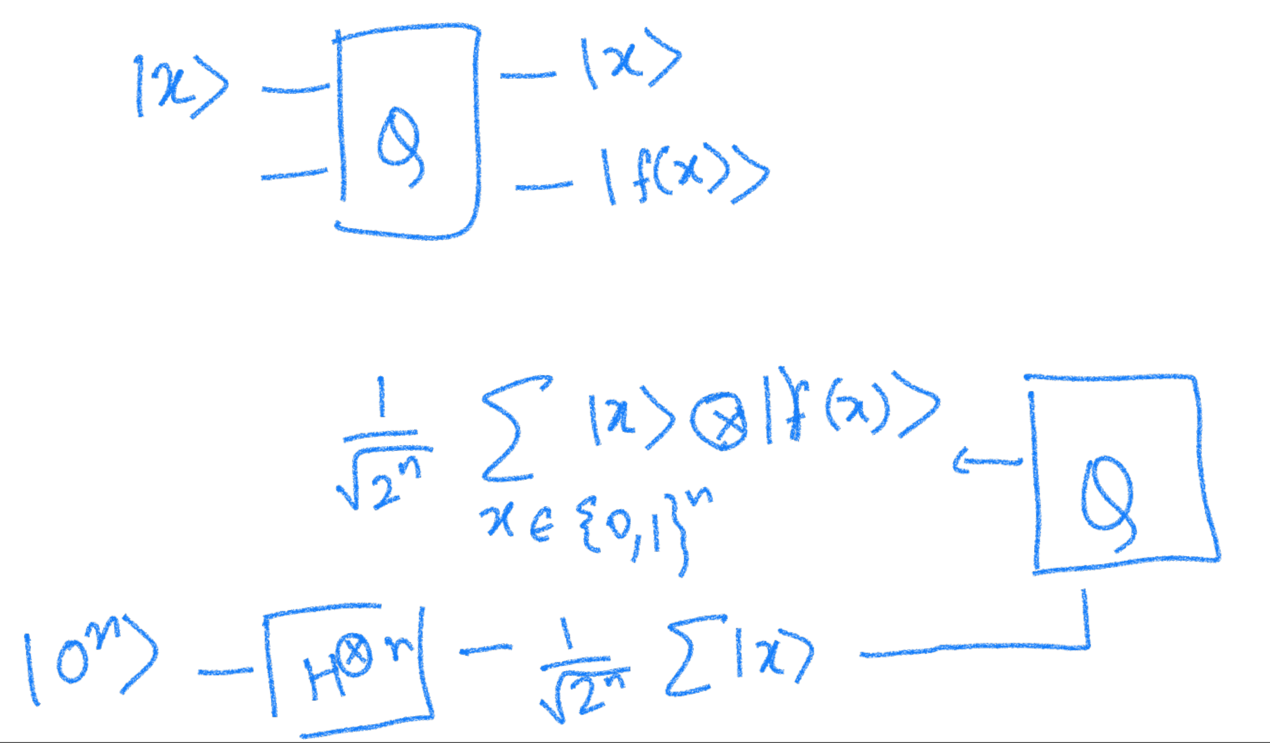
$$\frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} |x\rangle \otimes \left((-1)^{F(x)} |0\rangle \right)$$

$$= \frac{1}{\sqrt{N}} \sum_{x} (-1)^{F(x)} |x\rangle \otimes |0\rangle$$

$N = 2^n.$



$$\frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{F(x)} |x\rangle \otimes |0\rangle$$


$|x\rangle$
Q
$|x\rangle$
$|f(x)\rangle$
$\frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle \otimes |f(x)\rangle$
Q
$|0^n\rangle$
$H^{\otimes n}$
$\frac{1}{\sqrt{2^n}} \sum |x\rangle$

- So how do we use the power of quantum?
- Is there more that we can do with quantum gates?
- We will assume we can efficiently implement any 1/2/3 qubit transformation.

We want transformations that are

- easy to implement via simple gates/transformations
- Useful.

Q.C research is an interplay between these two goals.

- Sometimes we will get easy transformations and see how useful these are.
- With more experience, we will do a bit of both.

Possible simple transformations

$$|x_1 \ldots x_n\rangle \; — \; \boxed{H^{\otimes n}} \quad ?$$

Fourier transform over

$\{0,1\}^n$

or $\mathbb{Z}_2^n$

$$|0\rangle \xrightarrow{H} |+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{(-1)^0}{\sqrt{2}}|0\rangle$$

$$|1\rangle \xrightarrow{H} |-\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{(-1)^1}{\sqrt{2}}|1\rangle$$

$$|00\rangle \xrightarrow{H\otimes H} |++\rangle = \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \left(|0\rangle + |1\rangle \right) \otimes \left(|0\rangle + |1\rangle \right)$$

$$= \frac{1}{2}\left(|00\rangle + |01\rangle + |10\rangle + |11\rangle \right)$$

$$|01\rangle \xrightarrow{H\otimes H} |+-\rangle = \frac{1}{2}\left(|0\rangle + (-1)^0|1\rangle \right) \otimes \left(|0\rangle + (-1)^1|1\rangle \right)$$

$$= \frac{1}{2}\left(|00\rangle + (-1)^0|10\rangle + (-1)^1|01\rangle + (-1)^{1+0}|11\rangle \right.$$

$$|10\rangle \xrightarrow{H} \frac{1}{2}\left(|0\rangle + (-1)^1 |1\rangle\right) \otimes \left(|0\rangle + (-1)^0 |1\rangle\right)$$

$$\frac{1}{2}\left(|00\rangle + (-1)^0 |01\rangle + (-1)^1 |10\rangle + (-1)^{1+0} |11\rangle\right.$$

$$|11\rangle \xrightarrow{H} \frac{1}{2}\left(|0\rangle + (-1)^1 |1\rangle\right) \otimes \left(|0\rangle + (-1)^1 |1\rangle\right)$$

$$= \frac{1}{2}\, |00\rangle + (-1)^1 |01\rangle + (-1)^1 |10\rangle + (-1)^{1+1} |11\rangle$$

$$|b_1 b_2\rangle =$$

$$|b_1 b_2\rangle = \frac{1}{2}\left(|0\rangle + (-1)^{b_1}|1\rangle\right) \otimes \left(|0\rangle + (-1)^{b_2}|1\rangle\right)$$

$$= \frac{1}{2}\left(|00\rangle + (-1)^{b_2}|01\rangle + (-1)^{b_1}|10\rangle + (-1)^{b_1+b_2}|11\rangle\right)$$

$$\frac{1}{2} \sum_{\substack{x_1 \in \{0,1\} \\ x_2 \in \{0,1\}}} (-1)^{x_1 b_1 + x_2 b_2 (\text{mod} 2)} |x_1 x_2\rangle$$

$$|010\rangle \xrightarrow{H\otimes H\otimes H} |+-+\rangle = \frac{1}{\sqrt{8}}\left(|0\rangle+|1\rangle\right)\otimes\left(|0\rangle+(-1)^{1}|1\rangle\right)\otimes\left(|0\rangle+(-1)^{0}|1\rangle\right)$$

$$|101\rangle \xrightarrow{H^{\otimes 3}} |-+-\rangle = \frac{1}{\sqrt{2}}\left(\underline{\qquad\qquad} \quad \underline{\qquad} \right)$$

$$|b_1 b_2 b_3\rangle \xrightarrow{H^{\otimes 3}} \frac{\sum_{x\in\{0,1\}^3}(-1)^{b_1x_1+b_2x_2+b_3x_3 \pmod 2}\,|x_1x_2x_3\rangle}{\sqrt{8}}$$

$$|x\rangle = \begin{array}{c|c} 0 & 000 \\ 0 & 001 \\ 0 & 010 \\ 0 & 011 \\ 0 & 100 \\ 1 & 101 \\ 0 & 110 \\ 0 & 111 \end{array} \longrightarrow \frac{1}{\sqrt{8}}\begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \\ -1 \\ 1 \\ -1 \\ 1 \end{bmatrix} \to |X_x\rangle$$

$\to$ s^{th} co-ordinate $(-1)^{\sum x_i s_i \pmod 2}$

More generally

$$\boxed{x} \;—\; \boxed{H^{\otimes n}} \;—\; \boxed{X_x}$$

X_x

$$\begin{bmatrix} \\ \vec{s} \\ \\ \end{bmatrix}$$

$\vec{s}^{\,th}$ co-ordinate $(-1)^{\sum_{i=1}^{n} x_i s_i \pmod 2}$

$= |X_{\vec{x}}\rangle$